

WarfarinTwin

A GIFT-anchored digital twin for post-surgery warfarin dosing

Happiest Health Digital Twin Challenge 2026

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Why warfarin is hard



2 to 315

mg per week

stable dose range across 5,410 real patients (median 28)



INR 2 to 3

target range

too low risks clots, too high risks bleeding



INR $\geq$ 4

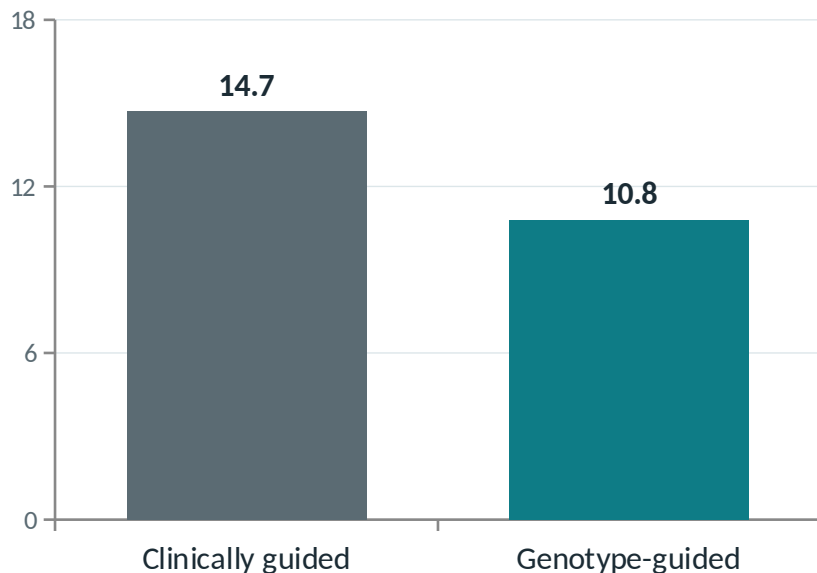
warning sign

bleeding risk climbs; this is what GIFT-style dosing tries to avoid

The right dose depends on genes (CYP2C9, VKORC1, CYP4F2), age, weight and interacting drugs, and the INR reacts with a delay of days. That delay makes trial-and-error dosing risky.

The clinical anchor: GIFT

Adverse events in the first 30 to 60 days (%)



1,650 patients

aged 65+, elective hip or knee replacement



Genotype-guided dosing

VKORC1, CYP2C9 and CYP4F2 used for the first 11 days



Mostly fewer INR ≥ 4

the benefit was driven by fewer high-INR episodes

Source: Gage et al., JAMA 2017;318(12):1115-1124

Our idea: a digital twin



1. Profile

age, weight, genes,
interacting drugs



2. Daily INR

one blood test each
morning



3. Update the twin

Bayesian update: how
sensitive is this
patient?

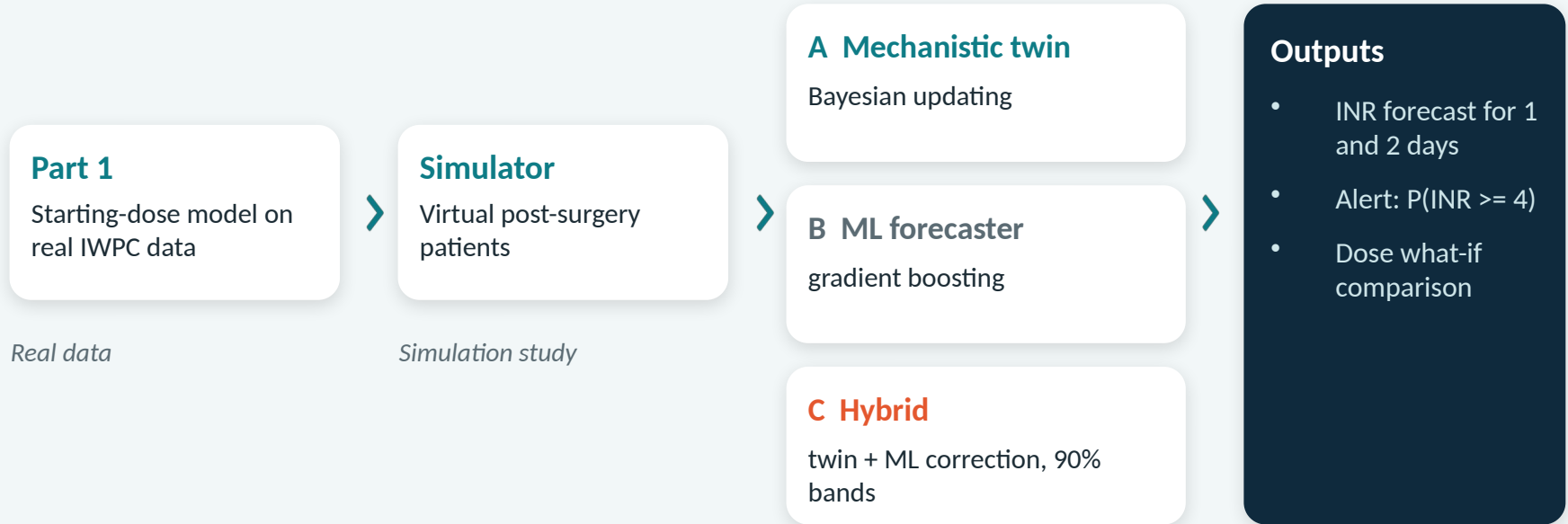


4. Forecast

INR tomorrow, alert
for INR ≥ 4 , dose
what-if

Like a flight simulator for a patient's dose: try a change on the virtual patient first, then on the real one.

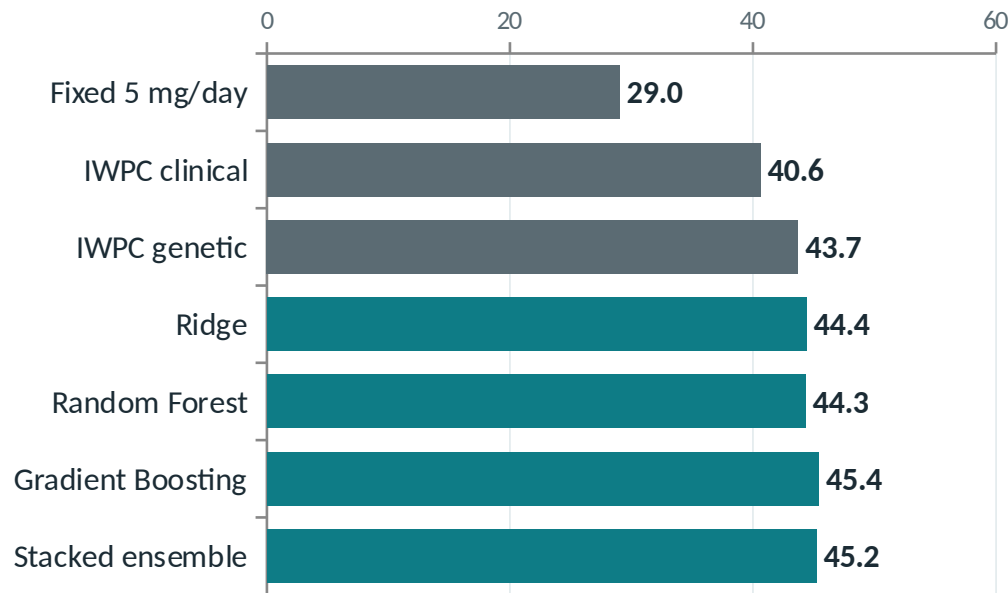
How the system fits together



The simulator's stable doses come from Part 1, so the two parts are linked: Part 1 gives the starting dose, the twin handles the days after.

Part 1: the starting dose

Patients whose predicted dose is within 20% of the true dose (%)



5,410

real IWPC patients

45%

within 20% of true dose, vs 29% for a flat 5 mg/day

VKORC1

most important feature, then age and CYP2C9

Honest note: ML improves only slightly over the published IWPC equation. Genotype matters more than the model.

Part 2: virtual patients



Hidden stable dose

Part 1 prediction times real-data variability



CYP2C9 sets elimination speed

slow metabolizers keep the drug longer



INR responds with a delay

about 2.5 days, so doses act late



Real-life noise

diet wobble, 8% start an interacting drug, test error

Models see

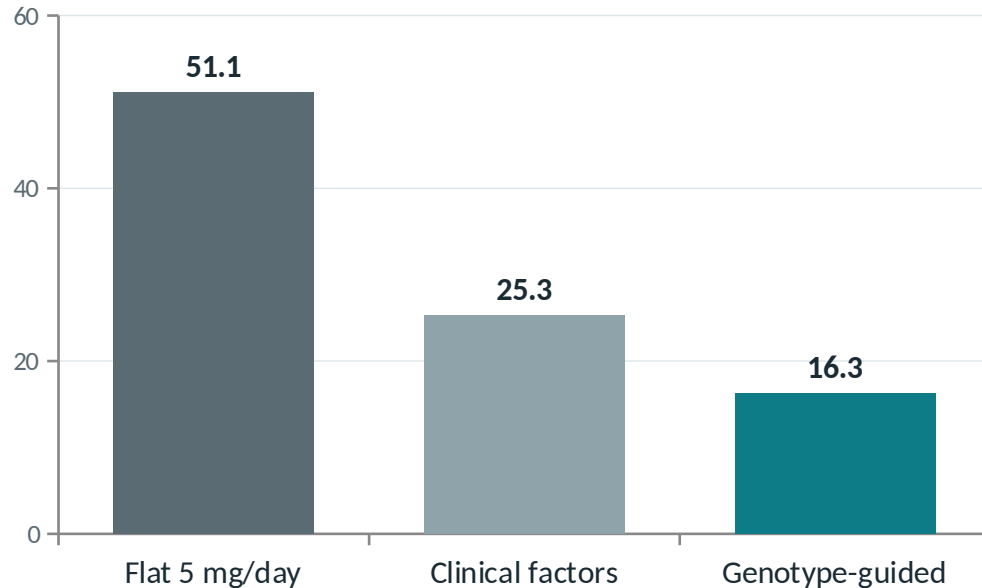
covariates, genotype (if tested),
doses given, daily INR

Models never see

true stable dose, elimination
speed, delay, wobble, interaction
events

Does a genotype start help?

Virtual patients with INR ≥ 4 at any point, days 1 to 11 (%)



36%

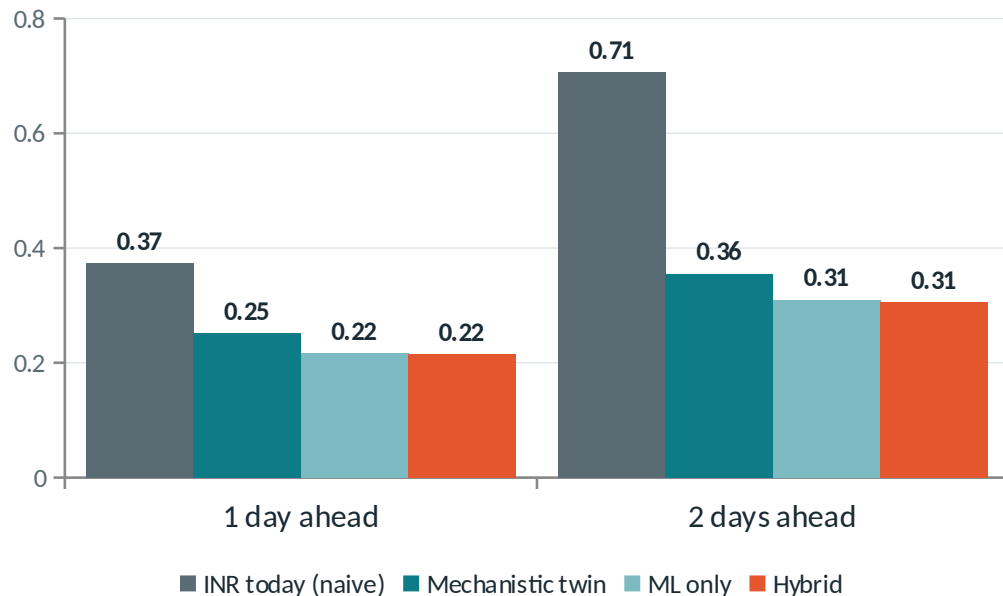
fewer INR ≥ 4 events with a genotype-guided start than a clinical-factors start

- Absolute drop 9.1 points (95% CI 7.1 to 11.1)
- Time in range 43.6% vs 38.6%
- Effect stays 28% to 43% under other assumptions

Same 1,500 virtual patients in every arm. Not tuned to GIFT: compare direction, not absolute rates.

Forecasting the next days

Average error of the INR forecast (lower is better)



0.215

INR average error for the hybrid next-day forecast (naive: 0.373)

0.98

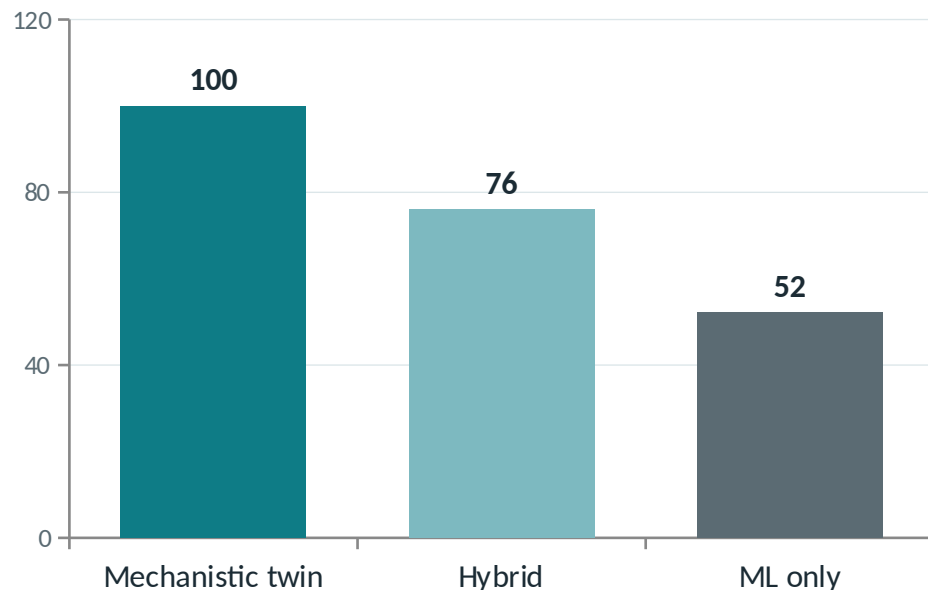
alert AUROC for INR ≥ 4 tomorrow (0.97 at two days)

90%

of true INRs fall inside the hybrid's 90% band (90.3% measured)

The key test: what if?

Predicts the right direction of a dose change's effect (%)



52% is a coin flip

Doctors hold the dose when INR is high, so pure machine learning learns 'less drug goes with high INR'. That is confounding by indication.

The twin models how the drug works, so it gets cause and effect right (correlation 0.97 with the true change).

Caveat: the twin shares the simulator's structure, so this is a best case.

One patient, five doses

Day 4 morning

- 72-year-old woman, 62 kg
- CYP2C9 *1/*3, VKORC1 A/A
- Given 2 mg each night so far
- INR rising: 1.0, 1.0, 1.1, 1.5, 2.3

Dose is repeated tomorrow night.

Dose tonight	Twin INR in 2 days	Chance INR ≥ 4
0 mg (hold)	3.26	11%
1 mg	3.70	31%
1.5 mg	3.92	44%
2 mg (same)	4.15	58%
3 mg	4.60	79%

Holding the dose tonight cuts the chance of INR ≥ 4 from 79% (3 mg) to 11%.

Honest limits



Simulation study

The twin shares the simulator's structure, so its accuracy is optimistic.



Our assumptions

CYP2C9 speeds, titration rules and variability are ours and flagged in the code.



Event rates differ

Absolute INR ≥ 4 rates are higher than GIFT's; read direction, not levels.



Dated data

IWPC is from 2008 and under-represents some populations.



Small ML gain

In Part 1, ML barely beats the published IWPC equation.



Not clinical

A research prototype, not validated on patients, not a medical device.

What comes next

1

Validate on real data

Check INR forecasts on real hospital INR time series (credentialed access needed)

2

Clinician dashboard

A simple Streamlit app around the dose what-if tool

3

Local genetics

India-relevant allele frequencies and explicit CYP4F2

4

Published parameters

Swap assumed values for published PK/PD estimates, with sensitivity tests

Step 1 is the most important: it is the evidence that would turn this prototype into something clinicians could trust.

Team and college



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Project lead and modeling. Third-year
B.Pharm student.



[Teammate name]

Data acquisition. [Year and course]



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Code and results: github.com/legallyananyar-dev/digital-twin-challenge-

References

Gage BF, et al. Effect of genotype-guided warfarin dosing on clinical events and anticoagulation control among patients undergoing hip or knee arthroplasty: the GIFT randomized clinical trial. JAMA. 2017;318(12):1115-1124.

International Warfarin Pharmacogenetics Consortium; Klein TE, et al. Estimation of the warfarin dose with clinical and pharmacogenetic data. N Engl J Med. 2009;360:753-764.

Johnson JA, et al. CPIC guideline for pharmacogenetics-guided warfarin dosing: 2017 update. Clin Pharmacol Ther. 2017;102(3):397-404.

Hamberg AK, et al. A PK-PD model for predicting the impact of age, CYP2C9, and VKORC1 genotype on individualization of warfarin therapy. Clin Pharmacol Ther. 2007 (structure inspired by; parameters not copied).

Data: Warfarin Consortium Combined Data Set (March 2008), IWPC / PharmGKB.